

Improving LatticeFold+ with ℓ_2 -norm Checks

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Abstract. Folding schemes enable incremental proving by compressing many relation instances into a small accumulator. Recent lattice-based constructions such as LatticeFold+ [CRYPTO '25] achieve post-quantum security, but prover performance is still dominated by expensive ℓ_∞ range checks used to control witness growth during folding and extraction. We present a final ℓ_2 -norm-check design that combines random-projection constraints (in the spirit of Rok and Roll [ASIACRYPT '25]) with an exact shortening step (in the spirit of SALSAA [ePrint 2025/2124]) to recover the original witness ℓ_2 -norm bound at extraction time. Integrated into the LatticeFold+ composition, this gives iterative folding with controlled norm growth, preserved binding and knowledge-soundness goals, and substantially lower prover cost on the dominant norm-check path, while maintaining a similar proof size and verification cost. Our approach is modular and can be applied to other lattice-based folding schemes, providing a practical path for efficient post-quantum folding constructions.

1 Introduction

In recent years, succinct non-interactive arguments of knowledge (SNARKs) have become central to applications such as blockchains [XZC⁺22, GKO24, LM25, LZW⁺25], privacy-preserving authentication [EHR24], and verifiable computation [ACGS24, CCC⁺25]. Despite their power, SNARKs suffer from a major memory bottleneck: the prover must load the entire computation into memory before proof generation can begin. This is infeasible for large computations and, more broadly, fails to capture how computations are carried out in practice, namely in a streaming manner.

This setting has led to the framework of *incrementally verifiable computation* (IVC) [Val08], in which a long-running computation is proven step by step. Early IVC protocols were built from SNARKs and can be viewed as producing, at each step, a proof for the computation performed by the verifier in the previous step. More modern approaches use *folding schemes* [BCL⁺21, BDFG21, BC25a, BC25b], which let the prover aggregate many computation steps into a single instance of an accumulator relation. Formally, a folding scheme realizes a reduction of knowledge [ACK21] from a product relation $\Xi_{\text{acc}} \times \Xi_0$ to an accumulation relation Ξ_{acc} : it folds a pair of instances (x_{acc}, x_0) into a single new instance x'_{acc} of Ξ_{acc} . Repeating this reduction allows the prover to incrementally aggregate many computation steps while maintaining only a constant-size accumulator state.

1.1 Lattice-based folding

The best-established folding schemes rely on linearly homomorphic commitments, such as Pedersen commitments [Ped92], which are vulnerable to quantum attacks [Sho94]. A major recent advance is the development of lattice-based folding schemes, including LatticeFold [BC25a], LatticeFold+ [BC25b], and Neo [NS25]. These schemes replace Pedersen commitments with Ajtai commitments [Ajt96], which are based on the hardness of lattice problems and are widely regarded as plausibly post-quantum secure.

However, translating Pedersen-based folding to the lattice setting is not straightforward: Ajtai commitments require the committed message to be *short*, i.e., the witness must have low norm. This shortness constraint creates difficulties because the witness norm typically grows with each folding step. To control this growth, all existing lattice-based folding schemes employ ℓ_∞ -norm

range-checks to certify that witness vectors satisfy prescribed bounds at each step. These range-checks constitute the dominant computational cost for the prover and represent the primary bottleneck in lattice-based folding. Put differently, the practicality of all currently known lattice-based folding schemes hinges on the witness norm being small enough to allow efficient ℓ_∞ -norm checks.

1.2 Our contribution

In this work, we show that the expensive ℓ_∞ -norm range-checks can be replaced by substantially more efficient ℓ_2 -norm checks. Our construction combines random-projection constraints, in the spirit of Rok and Roll [KLNO25], with an exact shortening step from SALSAA [KLOT25] to recover the original witness ℓ_2 -norm bound at extraction time. While we instantiate our techniques within the LatticeFold+ framework of Boneh and Chen [BC25b], the approach is modular and applies to other lattice-based folding schemes.

Concretely, we obtain the following improvements over the baseline LatticeFold+ construction:

1. The dominant prover cost on the norm-check (the numbers of additions in $\mathcal{R}_q$) is reduced by a magnitude and is now independent of the witness norm bound.
2. Security goals are preserved under the same (standard) lattice hardness assumptions.
3. Proof size and verification cost remain comparable to the baseline in the general parameter range.

1.3 Overview of Techniques

At a high level, we keep the LatticeFold+ architecture from [BC25b] and replace only the norm-control layer. The baseline uses an ℓ_∞ range-check pipeline based on monomial commitments; we replace it with an ℓ_2 pipeline built from two Reductions of Knowledge (RoKs; subprotocols between prover and verifier that transform one relation into another while preserving knowledge-soundness guarantees).

The first RoK is projection based (in the spirit of [KLNO25]). The prover block-projects each witness with a shared Johnson-Lindenstrauss matrix Π , commits to the projected image, and proves linear consistency with the original witness via sumcheck-friendly claims. The JL intuition is:

$$c_1 \|\mathbf{w}\|_2 \leq \|\Pi \mathbf{w}\|_2 \leq c_2 \|\mathbf{w}\|_2$$

with high probability (in our concrete lemma, $c_1 = \sqrt{30}$ and $c_2 = \sqrt{337}$). So low projected norm certifies low original norm up to a constant factor.

This step requires one crucial modification to folding: we append an image relation (for the projected witness) but we do *not* fold that image relation in the random-combination step. The reason is extraction. Random combination introduces slack (a short denominator from challenge differences). If the image relation were folded as well, its extracted witness would also carry slack and would no longer be a clean certificate for norm control of the original witness. Keeping the image relation unfolded preserves a slack-free handle for the extractor.

The second RoK is exact shortening (in the spirit of [KLOT25]). Projection gives a *shortish* witness; exact shortening removes the remaining constant-factor gap by enforcing

$$\text{ct}(\langle \mathbf{w}, \overline{\mathbf{w}} \rangle) \leq \beta^2,$$

where ct denotes the constant-term map on $\mathcal{R}_q$ (denotes the ring of integer polynomials modulo $X^N + 1$ with coefficients in $\mathbb{Z}_q$). Intuitively, this quantity is the squared ℓ_2 norm encoded in the ring inner product (for $N = 1$, i.e. $\mathcal{R}_q = \mathbb{Z}_q$, it is just $\langle \mathbf{w}, \mathbf{w} \rangle \leq \beta^2$). These are again reduced to evaluation claims compatible with the same homogenization/folding flow.

Composed with homogenization, random combination, and decomposition from [BC25b], this yields iterative folding without norm drift across rounds: projection gives shortish extraction, exact shortening restores the original target bound, and decomposition rebalances correctness-side growth. The dominant prover term on the norm-check path drops from $L \cdot n \cdot m \cdot N \cdot \log_N \beta$ ring additions to $L \cdot m \cdot 256$ additions (where L is the number of instances, $n \times m$ is the dimension of the commitment matrix and β is the norm bound on the witness), with similar proof-size and verifier-cost profiles in the reported range.

Due to the rather workshop-tailored nature of this publication, we do not provide formal security proofs as they follow verbatim from the similar proofs in [BC25b, KLOT25].

2 Notation

We denote by $[n]$ the set $\{0, 1, \dots, n-1\}$. Throughout this work, we use boldface letters to denote vectors and matrices, which are further denoted by lowercase and uppercase, respectively. We assume that vectors are column vectors. We write $\|\cdot\|$ to denote the norm of a vector, where the specific norm (e.g. ℓ_∞ or ℓ_2) is specified explicitly when needed. For concatenation of vectors, we use the notation $[\mathbf{a}||\mathbf{b}]$ to denote the concatenation of vectors $\mathbf{a}$ and $\mathbf{b}$.

We use $\mathcal{R}_q$ to denote the ring of integer polynomials modulo $X^N + 1$ with coefficients in $\mathbb{Z}_q$ for some integer modulus q and degree $N = 2^k$. For simplicity, one could set $\mathcal{R}_q = \mathbb{Z}_q$ (i.e. $N = 1$) to obtain standard integer vectors, which is mainly sufficient for our discussion. We use the notation $\otimes$ to denote the Kronecker product and $\odot$ to denote the Hadamard product.

Given a vector $\mathbf{v} \in \mathcal{R}_q^{2^\mu}$, its multi-linear extension is defined as the unique multi-linear polynomial $\text{MLE}[\mathbf{v}](X_0, \dots, X_{\mu-1})$ such that $\text{MLE}[\mathbf{v}](\mathbf{b}) = v_{\mathbf{b}}$ for every $\mathbf{b} \in \{0, 1\}^\mu$. If a vector has a structured form, i.e. it is a product of lower-dimensional vectors, then its multi-linear extension can be computed efficiently. We write $\mathcal{R}_q^{2^{\otimes \mu}}$ to denote the set of vectors $\mathbf{v} \in \mathcal{R}_q^{2^\mu}$ such that $\mathbf{v} = \mathbf{v}_0 \otimes \mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_{\mu-1}$ for some vectors $\mathbf{v}_i \in \mathcal{R}_q^2$.

We define a general (NP) relation Ξ as a triple:

$$\Xi := (\mathbf{pp}, \mathbf{stmt}, \mathbf{wtn}) \in \mathcal{P} \times \mathcal{S} \times \mathcal{W}$$

so that $\mathbf{pp}$ is the public parameter (fixed throughout the execution of the protocol), $\mathbf{stmt}$ is the statement and $\mathbf{wtn}$ is the witness. The relation holds if the predicate $R_\Xi : \mathcal{P} \times \mathcal{S} \times \mathcal{W} \rightarrow \{0, 1\}$ evaluates to 1; we write $(\mathbf{stmt}, \mathbf{wtn}) \in \Xi$ to denote that the instance-witness pair is valid.

A particular relation of interest is the (oversimplified) Rank-1 Constraint System (R1CS). R1CS is a widely used relation in the context of SNARKs and folding schemes, as many NP statements (including evaluations of arithmetic circuits) can be efficiently reduced to R1CS.

$$\Xi^{\text{R1CS}} := \left\{ \begin{array}{l} (\mathbf{A}, \mathbf{B}, \mathbf{C}), -, \mathbf{w} \\ \mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathcal{R}_q^{n \times n}, \quad \mathbf{w} \in \mathcal{R}_q^n \\ \mathbf{Aw} \odot \mathbf{Bw} = \mathbf{Cw} \bmod q, \end{array} \right\}$$

We remark that this arithmetization is considered here and also in [BC25b] for simplicity of exposition, but with some generic modifications (as in [BC25a]) we can extend our construction to support more advanced arithmetizations, e.g. customizable constraint systems [STW23], which further generalize arithmetizations such as AIR or Plonkish.

3 Boneh & Chen's Framework for Lattice-based Folding Schemes

3.1 Reductions of Knowledge

A reduction of knowledge (RoK) is an interactive algorithm (between prover and verifier) that takes an instance of a relation and produces another instance of a (possibly different) relation. We write $\Pi : \Xi_0 \rightarrow \Xi_1$ to denote a reduction from relation in Ξ_0 to relation in Ξ_1 .

We say that reduction of knowledge is correct (perfectly or statically) if for every valid instance-witness pair $(\text{stmt}_0, \text{wtn}_0) \in \Xi_0$, the output instance-witness pair $(\text{stmt}_1, \text{wtn}_1) = \Pi(\text{stmt}_0, \text{wtn}_0)$ is also valid, i.e. $(\text{stmt}_1, \text{wtn}_1) \in \Xi_1$. We also say that reduction of knowledge is knowledge-sound if for every statement $\text{stmt}_0 \in \Xi_0$ and for every valid witness wtn_1 for the output instance $(\text{stmt}_1, \text{wtn}_1) = \Pi(\text{stmt}_0, \text{wtn}_0)$, there exists an efficient extractor algorithm that can extract a valid witness wtn_0 for the input instance stmt_0 . The extractor is allowed to rewind the prover algorithm multiple times with the same input instance stmt_1 but different randomness.

The LatticeFold+ [BC25b] framework consists of the composition of *reductions of knowledge*.

3.2 Generalized Committed Linear Relation

Before proceeding to the description of the folding schemes, we also define the notion of *generalized committed linear relation*, which will be useful in the construction.

$$\Xi_\beta^{\text{lin}} := \left\{ \begin{array}{l} (\mathbf{A}, (\mathbf{M}_i)_{i \in [t]}), (\mathbf{y}, (v_i)_{i \in [t]}, \mathbf{r}), \mathbf{w} \\ \mathbf{A} \in \mathcal{R}_q^{n \times m}, \quad \mathbf{M}_i \in \mathcal{R}_q^{m \times m}, \\ \mathbf{y} \in \mathcal{R}_q^a, \quad v_i \in \mathcal{R}_q, \quad \mathbf{r} \in \mathcal{R}_q^{2^{\otimes \mu}}, \\ \mathbf{w} \in \mathcal{R}^m, \\ \begin{pmatrix} \mathbf{A} \\ \mathbf{r}^T \mathbf{M}_0 \\ \vdots \\ \mathbf{r}^T \mathbf{M}_{t-1} \end{pmatrix} \mathbf{w} = \begin{pmatrix} \mathbf{y} \\ v_0 \\ \vdots \\ v_{t-1} \end{pmatrix} \pmod{q}, \\ \|\mathbf{w}\| \leq \beta \end{array} \right\}.$$

In words, the public parameters consist of a matrix $\mathbf{A}$, which is viewed as an Ajtai-type commitment (binding under the ring variant of the short integer solution problem), and a set of matrices $(\mathbf{M}_i)_{i \in [t]}$, which form additional constraints on the witness. The statement consists of vectors $\mathbf{y}$ and $(v_i)_{i \in [t]}$ forming the right-hand side of the linear system (when $(\mathbf{M}_i)_{i \in [t]}$ are not squares, we can pad them with zeros without compromising the efficiency of the scheme). Further, the matrices $(\mathbf{M}_i)_{i \in [t]}$ are “batched” via vector $\mathbf{r}$ so that the right-hand side remains succinctly represented. The witness $\mathbf{w}$ is required to satisfy the linear system and also to be of short norm (in the context of [BC25b], the ℓ_∞ -norm is used).

3.3 Folding Scheme for R1CS Relation

Eventually, we are ready to describe the folding scheme.

The folding scheme for the R1CS relation is obtained with two main ingredients:

1. Linearization reduction from the R1CS relation to the generalized committed linear relation,
2. Folding scheme for the generalized committed linear relation.

The latter, in turn, is obtained via the composition of the following reductions of knowledge:

- Norm check reduction that forms a “checkpoint” on the norm of the witness so that the extracted witness has the correct norm.
- Homogenization that makes the left-hand side of the relations uniform (i.e. the same vector $\mathbf{r}$ across all instances).
- Random linear combination of witnesses and right-hand sides to reduce the number of instances from L to 1.
- Decomposition that reduces the norm of the witness by splitting it into two parts, forming two new instances of the linear relation.

The objective of this complex sequence of reductions is to obtain a folding scheme that reduces the number of instances from L to 2 while keeping the norm of the witness under control. In more detail, we expect that during folding (i.e. in the *correctness* direction) and during extraction (i.e. in the *knowledge-soundness* direction), the norm of the witness does not grow. This restriction is extremely important, as otherwise, after a few folding steps, the norm of the witness would become too large for the Ajtai-style commitment to remain binding.

In the correctness direction, the norm of the witness grows during the combination step, but is reduced back via decomposition. In the knowledge-soundness direction, the norm of the witness grows during decomposition. In the folding step, the norm not only grows during the extraction, but the witness is also augmented with a “short denominator” (also known as slack). Such a relation, often denoted *relaxed*, remains binding but with a significant parameter degradation¹. This temporary growth must be controlled, which is the role of the norm-check step. The norm-check step ensures that the norm of the extracted witness is precisely the same as the norm of the original witness.

In this work, we will ignore the linearization reduction, as it is not relevant for our discussion (we refer the reader to [BC25a] for details). We will focus on the folding scheme for the generalized committed linear relation.

3.4 Norm-check RoK

The norm-check reduction is a reduction of knowledge from the generalized committed linear relation to the same relation with additional constraints. The reduction serves as a “checkpoint” on the norm of the witness. In the extraction direction, it ensures that the extracted witness has the same norm as the original witness.

In a nutshell, the reduction works as follows: the witnesses of the input relation are decomposed into monomials, i.e. for each witness $\mathbf{w} \in \mathcal{R}_q^m$, we write $\mathbf{w} = \sum_j \mathbf{w}^{(j)} X^j$. Then, the prover commits to each monomial vector via Ajtai-style commitments and proves consistency between the committed monomials and the original witness. Therefore, since the degree of the polynomial is bounded, the prover argues that the norm of the original witness is bounded via the ℓ_∞ norm.

To save communication, the prover does not send the commitments to all monomials, but instead employs a “double-commitment” technique. Nevertheless, the computational overhead remains significant. To compute the commitments to all monomials, the prover needs to perform $L \cdot n \cdot m \cdot N \cdot \log_N \beta$ additions of cyclotomic ring elements, where L is the number of instances, $n \times m$ is the dimension of the commitment matrix and N is the degree of the ring. This overhead might be prohibitive in practice.

3.5 Homogenization RoK

The homogenization reduction is a reduction of knowledge from the generalized committed linear relation with additional constraints to itself. The reduction ensures that the left-hand side of the relation is uniform across all instances, i.e. the vector $\mathbf{r}$ is the same for all instances.

The prover and verifier express all linear relations in terms of sumcheck claims, i.e., for each instance $j \in [L]$, they express the relation as

$$\sum_{\mathbf{b} \in \{0,1\}^\mu} \text{MLE}[\mathbf{r}_j](\mathbf{b}) \cdot \text{MLE}[\mathbf{M}_i \mathbf{w}_j](\mathbf{b}) = y_{i,j}$$

¹ To get an intuition about the “slack”, consider a simple case of folding two ISIS instances, i.e. $\mathbf{A}\mathbf{w}_0 = \mathbf{y}_0$ and $\mathbf{A}\mathbf{w}_1 = \mathbf{y}_1$, into $\mathbf{A}\mathbf{v} = \mathbf{u}$, where $\mathbf{v} = \mathbf{w}_0 + c \cdot \mathbf{w}_1$ and $\mathbf{u} = \mathbf{y}_0 + c\mathbf{y}_1$. To “undo this folding”, we need two transcripts with $(\mathbf{u}_0, \mathbf{v}_0)$ and $(\mathbf{u}_1, \mathbf{v}_1)$ such that $\mathbf{u}_0 = \mathbf{y}_0 + c_0\mathbf{y}_1$ and $\mathbf{u}_1 = \mathbf{y}_0 + c_1\mathbf{y}_1$. Then we can extract $\mathbf{w}_0$ and $\mathbf{w}_1$ as $\mathbf{w}_0 = \frac{c_1\mathbf{v}_0 - c_0\mathbf{v}_1}{c_1 - c_0}$ and $\mathbf{w}_1 = \frac{\mathbf{v}_0 - \mathbf{v}_1}{c_1 - c_0}$. The denominator $c_1 - c_0$ is the slack. This intuition extends to the more general case of folding the generalized committed linear relation, as well as to the case of folding more than two instances, as explained e.g. in [FMN24, BC25b, KLNO25].

for each $i \in [t]$. Then, the prover and verifier engage in a sumcheck protocol² (batching all claims) to reduce the sumcheck claims to evaluation claims over a single random point $\mathbf{r}^* \in \mathcal{R}_q^\mu$, obtaining the following set of claims (claims over $\text{MLE}[\mathbf{r}_j](\mathbf{r}^*)$ are verified by the verifier directly):

$$\text{MLE}[\mathbf{M}_i \mathbf{w}_j](\mathbf{r}^*) = v_{i,j}^*, \quad \forall i \in [t], j \in [L].$$

where $v_{i,j}^*$ are appropriately defined linear combinations of the original right-hand sides. These can be expressed in matrix-vector form as:

$$\mathbf{r}^* \mathbf{M}_i \mathbf{w}_j = v_{i,j}^*, \quad \forall i \in [t], j \in [L].$$

Observe that the new Ξ^{lin} relations have uniform left-hand side $\mathbf{r}^*$ across all instances:

$$\begin{pmatrix} \mathbf{A} \\ \mathbf{r}^{*\text{T}} \mathbf{M}_0 \\ \vdots \\ \mathbf{r}^{*\text{T}} \mathbf{M}_{t-1} \end{pmatrix} \mathbf{w}_j = \begin{pmatrix} \mathbf{y}_j \\ v_{0,j}^* \\ \vdots \\ v_{t-1,j}^* \end{pmatrix} \pmod{q} \quad \forall j \in [L]. \quad (1)$$

3.6 Combination/Folding and Decomposition RoKs

The folding reduction is a reduction of knowledge from a list of generalized committed linear relation instances to a single instance of the same relation. The reduction randomly combines L instances into one instance.

Following the observation that in 1, the left-hand side of the relation is uniform across all instances, we can randomly combine the right-hand sides of the relations to obtain a single instance:

$$\begin{pmatrix} \mathbf{A} \\ \mathbf{r}^{*\text{T}} \mathbf{M}_0 \\ \vdots \\ \mathbf{r}^{*\text{T}} \mathbf{M}_{t-1} \end{pmatrix} \mathbf{v} = \begin{pmatrix} \mathbf{y} \\ v_0 \\ \vdots \\ v_{t-1} \end{pmatrix} \pmod{q}$$

where $\mathbf{v} = \sum_{j \in [L]} c_j \mathbf{w}_j$, $\mathbf{y} = \sum_{j \in [L]} c_j \mathbf{y}_j$ and $v_i = \sum_{j \in [L]} c_j v_{i,j}^*$ for each $i \in [t]$, for some random (short) coefficients c_j .

In the decomposition, the prover splits the witness into two parts, so that the norm of each part is reduced by a factor of two. In more detail, it takes an instance with witness $\mathbf{w}$ and produces two instances with witnesses $\mathbf{w}_0$ and $\mathbf{w}_1$ such that $\mathbf{w} = \mathbf{w}_0 + \rho \mathbf{w}_1$ for some base ρ . It sends the right-hand sides of the new instances to the verifier, who checks that they are consistent with the original instance.

4 Replacing Range-Check with ℓ_2 -Norm-Check

To mitigate the overhead of the norm-check reduction, we propose replacing the ℓ_∞ -norm check with an ℓ_2 -norm check.

4.1 Johnson-Lindenstrauss Lemma

The Johnson-Lindenstrauss lemma states that a small set of points in a high-dimensional space can be embedded into a lower-dimensional space such that pairwise distances are nearly preserved. In cryptographic applications, a simpler formulation is often more useful. Following [BS23], we state the lemma as follows:

² We remark that the sumcheck over $\mathcal{R}_q$ does not involve communication of “full” ring elements. We exploit the fact that $\mathcal{R}_q \cong (\mathbb{F}_{q^e})^{N/e}$. We set q^e to be superpolynomial (in the security parameter). Now, executing sumcheck over $\mathcal{R}_q$ is viewed as N/e parallel sumchecks over $\mathbb{F}_{q^e}$. Those sumchecks are batched.

Lemma 1 (Lemma 4.1 (Corollary 3.2, [GHL22]) with Corollary 4.2 [BS23]). *Let C be a distribution on $\{-1, 0, 1\}$ with $\Pr[C = 0] = 1/2$ and $\Pr[C = 1] = \Pr[C = -1] = 1/4$. Then, for every vector $\mathbf{w} \in \mathbb{Z}^d$ such that $\|\mathbf{w}\|_2 \leq q/125$, we have*

$$\sqrt{30}\|\mathbf{w}\|_2 < \|H\mathbf{w} \bmod q\|_2 < \sqrt{337}\|\mathbf{w}\|_2.$$

This holds except with probability 2^{-128} over the choice $H \leftarrow C^{256 \times d}$.

In words, the lemma states that by projecting a high-dimensional vector onto a lower-dimensional space using a random matrix from the specified distribution, the projected ℓ_2 norm remains within constant factors of the original ℓ_2 norm with overwhelming probability. The lemma is specified over the integers, but it can be easily adapted to the cyclotomic ring setting by embedding each entry of the projection matrix H as a constant polynomial.

4.2 Attempt I: [BS23]-style ℓ_2 -Norm-Check

We first consider a straightforward adaptation of the norm-check reduction from [BS23], which employs the Johnson-Lindenstrauss lemma to perform ℓ_2 -norm checks.

We consider the following reduction of knowledge from the generalized committed linear relation to a similar relation with additional constraints:

RoK $H^{\text{proj-BS23}}$:

- The verifier samples a random matrix $H \leftarrow C^{256 \times m}$ from the distribution specified in Lemma 4.1. We implicitly lift the matrix H from integers to cyclotomic ring elements by embedding each entry as a constant polynomial.
- The prover computes the projected witness $\mathbf{v}_j = H\mathbf{w}_j \bmod q$ for each instance $j \in [L]$ and sends it to the verifier.
- The verifier checks that the projected witness satisfies the norm constraints. The verifier sends a challenge $\mathbf{c} \in \mathcal{R}_q^{256}$ to the prover.

Later, in the homogenization step, both the prover and verifier express the relations as a sumcheck claim for each instance $j \in [L]$:

$$\sum_{\mathbf{b} \in \{0,1\}^\mu} \text{MLE}[\mathbf{c}^T H](\mathbf{b}) \cdot \text{MLE}[\mathbf{w}_j](\mathbf{b}) = \mathbf{c}^T \mathbf{v}_j \bmod q.$$

These relations are turned into evaluation claims over a single random point $\mathbf{r} \in \mathcal{R}_q^\mu$, as before. The claim over $\text{MLE}[\mathbf{c}^T H](\mathbf{r})$ is verified directly by the verifier. This approach has two disadvantages:

- The verifier needs to sample a new random matrix H and further compute evaluations of $\text{MLE}[\mathbf{c}^T H](\mathbf{b})$ during the homogenization step. This introduces significant overhead for the verifier, which is linear in the witness dimension m and is typically prohibitive.
- The extraction guarantee is weakened. Although the Johnson-Lindenstrauss lemma ensures that with overwhelming probability the norm of the projected witness is concentrated around the norm of the original witness (with some scaling factor), the bound holds only up to a (very small) constant factor. This means that during extraction, the norm of the extracted witness can be larger than the original witness by a constant factor. Repeated application of this reduction during multiple folding steps can lead to unbounded growth of the witness norm, eventually breaking the binding property of the commitments.

4.3 Attempt II: [KLNO25]-style ℓ_2 -Norm-Check

To address the verifier efficiency issue identified in the previous approach, we propose an alternative ℓ_2 -norm-check reduction inspired by the techniques from [KLNO25].

The idea is simple: instead of projecting the witness with a single projection matrix, we use multiple projections with shared randomness. In other words, we project the witness in small blocks so that each block is projected independently. Then, instead of sending a single projected witness, the prover commits to all projected blocks. In more detail, we consider the following reduction of knowledge from the generalized committed linear relation to a similar relation:

Assume L is a power of two and let $b = 256 \cdot L$ be the block size.

RoK $\Pi^{\text{proj-KLNO25}}$:

- The verifier samples a random matrix $\Pi \leftarrow C^{256 \times b}$ from the distribution specified in Lemma 4.1, where b is the block size. We implicitly lift the matrix Π from integers to cyclotomic ring elements by embedding each entry as a constant polynomial.
- For each $j \in [L]$, the prover splits witness $\mathbf{w}_j \in \mathcal{R}_q^m$ into m/b blocks of size b , i.e. $\mathbf{w}_j = [\mathbf{w}_j^{(0)} || \mathbf{w}_j^{(1)} || \dots || \mathbf{w}_j^{(m/b-1)}]$. The prover computes the projected blocks $\mathbf{v}_j^{(i)} = \Pi \mathbf{w}_j^{(i)} \bmod q$ for each block. It combines the projected blocks into $\mathbf{v}_j = [\mathbf{v}_j^{(0)} || \mathbf{v}_j^{(1)} || \dots || \mathbf{v}_j^{(m/b-1)}]$ and then forms $\mathbf{v} = [\mathbf{v}_0 || \mathbf{v}_1 || \dots || \mathbf{v}_{L-1}]$. The prover commits to $\mathbf{v}$ with $\mathbf{A}$ and sends the commitment to the verifier (forming a new instance of Ξ^{lin} for $\mathbf{v}$).
- The verifier sends a challenge $\mathbf{c} \in \mathcal{R}_q^{\mu \otimes \mu}$ to the prover, which is split into $\mathbf{c}^{(0)} \in \mathcal{R}_q^{2 \otimes \mu - e}$ and $\mathbf{c}^{(1)} \in \mathcal{R}_q^{2 \otimes e}$ so that $\mathbf{c} = \mathbf{c}^{(0)} \otimes \mathbf{c}^{(1)}$.
- The prover sends $s \in \mathcal{R}_q$ and $\mathbf{t} \in \mathcal{R}_q^L$ such that

$$\mathbf{c}_1^T (\mathbf{I}_{m/b} \otimes \Pi) \mathbf{w}_j = t_j \bmod q$$

$$(\mathbf{c}_0^T \otimes \mathbf{c}_1) \mathbf{v} = s \bmod q$$

- The verifier checks that $\sum_{j \in [L]} c_{0,j} t_j = s$.

Later, in the homogenization step, both the prover and verifier express the relations as a sumcheck claim for each instance $j \in [L]$, where t_j are the appropriately defined right-hand sides:

$$\sum_{\mathbf{b} \in \{0,1\}^\mu} \text{MLE}[\mathbf{c}_1 (\mathbf{I}_{m/b} \otimes \Pi)](\mathbf{b}) \cdot \text{MLE}[\mathbf{w}_j](\mathbf{b}) = t_j \bmod q.$$

$$\sum_{\mathbf{b} \in \{0,1\}^\mu} \text{MLE}[\mathbf{c}_0 \otimes \mathbf{c}_1](\mathbf{b}) \cdot \text{MLE}[\mathbf{v}](\mathbf{b}) = s \bmod q.$$

Those claims are turned into evaluation claims over a single random point $\mathbf{r} \in \mathcal{R}_q^\mu$, as before. Claims over $\text{MLE}[\mathbf{c}_1 (\mathbf{I}_{m/b} \otimes \Pi)](\mathbf{r})$ and $\text{MLE}[\mathbf{c}_0 \otimes \mathbf{c}_1](\mathbf{r})$ are verified directly by the verifier.

We further modify the protocol. To use the newly appended witness $\mathbf{v}$ in extraction, we need to ensure that this image has low norm. Otherwise, extraction might fail to produce a valid witness for the original relation. In other words, the relation corresponding to $\mathbf{v}$ must not be *relaxed*. To achieve this, we consider a slightly different folding variant that does not combine the projection witness. Therefore, the folding step randomly combines only L out of $L+1$ instances, leaving the relation corresponding to $\mathbf{v}$ untouched. This way, during extraction, the projected witness $\mathbf{v}$ is available to the extractor without slack, and the extractor uses the relation for $\mathbf{v}$ to argue that the projected witness has low norm via Lemma 1.

The output of folding is now a pair of two instances: the first instance corresponds to the randomly combined L instances, while the second instance corresponds to the relation for $\mathbf{v}$.

After decomposition, each is split into two instances, yielding four instances in total, which serve as an accumulator for the next folding step.

The proposed protocol is more efficient than the previous approach (given in 4.2), as computing $\text{MLE}[\mathbf{c}_1(\mathbf{I}_{m/b} \otimes \Pi)](\mathbf{r})$ is simplified by the mixed-product property of the Kronecker product.

However, it still leaves the extraction guarantee slightly open. In particular, we can “upgrade” the extracted witness norm from relaxed to almost the original norm. Nevertheless, the projection technique still leaves a small constant-factor gap between the norm of the original witness and the extracted witness. Repeated application of this reduction during multiple folding steps can lead to unbounded witness growth, eventually breaking the binding property of the commitments.

4.4 Final Attempt: From Shortish to Short Witnesses

To completely close the extraction guarantee gap, we combine the ℓ_2 -norm check with a final “shortening” step, which transforms a shortish witness into a short witness. This idea was first explored in [KLOT25] in the context of SNARKs.

This time, we exploit the fact that the witness is already known (from the extraction point of view) to be shortish, i.e. its ℓ_2 -norm is bounded by $\alpha\beta$, where β is the original witness norm and α is the factor by which the norm is increased during folding. In other words, we assume that the RoK $\Pi^{\text{proj-KLNO25}}$ is applied as a subroutine in the folding scheme, and we are now at the extraction point where we have a witness that is shortish but not necessarily short. We want to show that we can still argue that the witness is actually short, i.e. its ℓ_2 -norm is at most β .

We make the following observation: the ℓ_2 -norm of a vector is closely related to its self-inner product. In more detail, we write

$$\|\mathbf{w}\|_2^2 = \text{ct}(\langle \mathbf{w}, \overline{\mathbf{w}} \rangle)$$

where $\overline{\mathbf{w}}$ is the coordinate-wise complex conjugate of $\mathbf{w}$. In the case of power-of-two cyclotomic rings, complex conjugation corresponds to the automorphism $X \mapsto -X^{N-1}$. For further details, we refer the reader to [KLNO24]. We write $\text{ct} : \mathcal{R}_q \rightarrow \mathbb{Z}_q$ to denote the constant term of the cyclotomic ring element. We note that since $\|\mathbf{w}\|_2^2$ is already known to be small (i.e. less than $\alpha^2\beta^2$), the equality above holds over the integers (i.e. without mod q) for a sufficiently large modulus.

Therefore, to ensure that the witness is exactly of the desired norm, it suffices to show that $\text{ct}(\langle \mathbf{w}, \overline{\mathbf{w}} \rangle) = u \bmod q$ for some u s.t. $u \leq \beta^2$.

To achieve this, we consider the following reduction of knowledge from the generalized committed linear relation to a similar relation with additional constraints:

RoK $\Pi^{\text{exact-KLOT25}}$:

- The prover computes the inner product $u_j = \langle \mathbf{w}_j, \overline{\mathbf{w}}_j \rangle$ for each $j \in [L]$ and sends it to the verifier.
- The verifier checks that $\text{ct}(u_j) \leq \beta^2$ for each $j \in [L]$.
- The prover and verifier express the relations as a sumcheck claim for each instance $j \in [L]$, where u_j are the right-hand sides:

$$\sum_{\mathbf{b} \in \{0,1\}^\mu} \text{MLE}[\mathbf{w}](\mathbf{b}) \cdot \text{MLE}[\overline{\mathbf{w}}](\mathbf{b}) = u_j \bmod q.$$

- The prover and verifier engage in a sumcheck protocol and reduce the claims to a single random point $\mathbf{r}^* \in \mathcal{R}_q^\mu$, obtaining the following set of equations:

$$\text{MLE}[\mathbf{w}_j](\mathbf{r}^*) \cdot \text{MLE}[\overline{\mathbf{w}}_j](\mathbf{r}^*) = u_j^* \bmod q, \quad \forall j \in [L].$$

- The prover splits the equations into two linear equations and sends u'_j and u''_j so that for each $j \in [L]$:

$$\text{MLE}[\mathbf{w}_j](\mathbf{r}^*) = u'_j \bmod q$$

$$\overline{\text{MLE}[\overline{\mathbf{w}_j}](\mathbf{r}^*)} = \text{MLE}[\mathbf{w}_j](\mathbf{r}^*) = u_j'' \bmod q.$$

- The verifier checks that $u_j' \cdot \overline{u_j''} = u_j^*$ for each $j \in [L]$.

Later, in the homogenization step, both the prover and verifier express the evaluation claims over $\mathbf{r} \in \mathcal{R}_q^\mu$ (and $\mathbf{r}^* \in \mathcal{R}_q^\mu$), i.e., for each instance $j \in [L]$:

$$\sum_{\mathbf{b} \in \{0,1\}^\mu} \text{MLE}[\mathbf{w}_j](\mathbf{b}) \cdot \text{eq}(\mathbf{b}, \mathbf{r}) = u_j' \bmod q.$$

Those claims are then turned into evaluation claims over a single random point³.

The aforementioned reduction ensures that during extraction, the extracted witness has exactly the same norm as the original witness, provided that the extracted witness is already guaranteed to be shortish. Therefore, by combining this reduction with the Johnson-Lindenstrauss-based ℓ_2 -norm check, we obtain a complete ℓ_2 -norm-check reduction with an exact extraction guarantee.

Folding Scheme with ℓ_2 -Norm Check We now spell out the composition of the folding scheme with the ℓ_2 -norm-check reduction. The folding scheme consists of the following sequence of reductions of knowledge:

$$(\Xi_\beta^{\text{lin}})^L \xrightarrow{\Pi^{\text{exact-KLOT25}}} \underset{+ \text{ eval. constr.}}{(\Xi_\beta^{\text{lin}})^L} \xrightarrow{\Pi^{\text{proj-KLN025}}} \underset{+ \text{ proj. constr.}}{(\Xi_{\beta'}^{\text{lin}})^L} \xrightarrow{\Pi^{\text{hom+comb}}} (\Xi_{\beta''}^{\text{lin}})^2 \xrightarrow{\Pi^{\text{decomp}}} (\Xi_{\beta'''}^{\text{lin}})^4,$$

where $\beta < \beta' < \beta'' > \beta'''$ are the witness norm bounds at each step, and the additional constraints are as described in the respective reductions of knowledge. With appropriate parameter selection, we can ensure that $\beta''' \leq \beta$, so the final output of the folding scheme is a tuple of four instances of the linear relation with the same norm bound as the original witness. This allows the folding scheme to be applied iteratively without unbounded growth of the witness norm. The final output tuple serves as the accumulator for the next folding step.

The norm-related reductions of knowledge, $\Pi^{\text{exact-KLOT25}}$ and $\Pi^{\text{proj-KLN025}}$, are described in the previous sections. The homogenization-and-combination reduction $\Pi^{\text{hom+comb}}$ and the decomposition reduction Π^{decomp} are as described in [BC25b].

During extraction, the witness norm grows while executing the extractor for Π^{decomp} . In addition, the extractor for $\Pi^{\text{hom+comb}}$ extracts only a witness for the relaxed relation, which introduces slack in the norm bound. However, the extractor for $\Pi^{\text{proj-KLN025}}$ ensures that the extracted witness is shortish, and then the extractor for $\Pi^{\text{exact-KLOT25}}$ ensures that the extracted witness is actually short, i.e., it has the same norm bound as the original witness.

Remark 1. RoKs $\Pi^{\text{proj-KLN025}}$ and $\Pi^{\text{exact-KLOT25}}$ are presented as being applied independently to each Ξ^{lin} instance for expository clarity. For concrete efficiency, however, it is more efficient to batch them across all L instances. In particular, the projection matrix Π can be shared across instances, and the inner-product sumcheck in $\Pi^{\text{exact-KLOT25}}$ can also be batched. If the RoKs are applied independently per instance (without batching), then during homogenization the prover must send “cross-terms” for the projections and evaluation claims, which introduces communication overhead.

³ An alternative approach is to not execute the sumcheck protocol inside $\Pi^{\text{exact-KLOT25}}$, but instead also batch this sumcheck with the other sumcheck claims during the homogenization step. Then, the output of the homogenization step would include the evaluation claims for $\text{MLE}[\mathbf{v}](\mathbf{r}^*)$ and $\text{MLE}[\mathbf{v}](\mathbf{r}^*)$.

Metric	Baseline (LatticeFold+ [BC25b])	This work (ℓ_2 norm check)
Dominant prover term (norm-check path)	$L \cdot n \cdot m \cdot N \cdot \log_N \beta$ ring additions (monomial commitments)	$L \cdot m \cdot 256$ ring additions (JL projections)
Verifier cost	Verifier-efficient design from [BC25b]; includes ℓ_∞ norm-check verification	No additional asymptotic blow-up over the [BC25b] composition; the heaviest verifier computation is evaluating $\text{MLE}[\mathbf{c}_1 (\mathbf{I}_{m/b} \otimes \mathbf{H})](\mathbf{r})$ (magnitude of 1 ms for concrete parameters)
Proof size (reported/estimated)	About 100 KB for the matched setting ($L = 3$), as reported in [BC25b]	About 110 KB for $L = 5$, or about 175 KB for $L = 8$
Extraction norm guarantee	Norm control enforced by the ℓ_∞ range-check via monomial set check [BC25b]	After $\Pi^{\text{proj}}\text{---KL025}$ and $\Pi^{\text{exact}}\text{---KL025}$, extraction returns a witness with the original ℓ_2 target bound

Table 1. Compact comparison between baseline LatticeFold+ and the proposed ℓ_2 -norm-check variant.

5 Concrete Parameters and Efficiency

From an efficiency perspective, the proposed ℓ_2 -norm check reduction is significantly more efficient for the prover than the original ℓ_∞ -norm check from [BC25b]. The dominant step is now Johnson-Lindenstrauss projection computation, requiring $L \cdot m \cdot 256$ ring-element additions. This is an order of magnitude more efficient than the monomial-commitment step in [BC25b], where the number of additions is $L \cdot n \cdot m \cdot N \cdot \log_N \beta$ (for concrete parameters, n is typically set to $\log m / \log \lambda$). Moreover, these additions are over low-norm elements, so they can often be implemented without modular reductions and vectorized efficiently due to short coefficient bit-length.

Table 1 summarizes the high-level comparison with the baseline LatticeFold+ design.

The proof size has been estimated (see supplemental notebook at [GitHub](#)) for the following parameters: $L = 8$, $q = 2^{64}$, $N = 128$, $m = 2^{20}$, $B = 2^{25}$, and $n = 12$. For these parameters, the proof size is estimated at around 175 KB. The settings are chosen to almost match [BC25b], except for a smaller modulus q and a larger norm bound (because we use ℓ_2 norms). Further, in [BC25b], the number of instances is $L = 3$ (two accumulator instances and one new-relation instance). Here we set $L = 8$ (four accumulator instances and four new-relation instances), which is convenient because L is then a power of two. Adjusting to $L = 5$ to match [BC25b] (four accumulator instances and one new-relation instance) gives an estimated proof size around 110 KB, matching the numbers from [BC25b].

6 Towards a Tighter Analysis and Further Extensions

For simplicity of exposition, we omitted some details regarding precise parameter selection and a tighter analysis of the soundness guarantee. We also limited our discussion to the case of power-of-two cyclotomic rings and to the ring variant of the RICS relation. We briefly discuss these points below.

1. We ignored the impact of the union bound on proof soundness, i.e. we did not account for how many times the Johnson-Lindenstrauss lemma is applied across multiple folding steps. This can be fixed by adjusting lemma parameters accordingly, possibly increasing proof size in a negligible way.
2. In the efficiency analysis, we considered only the dominant prover term, namely Johnson-Lindenstrauss projections. In practice, other steps, especially homogenization shall not be ignored, particularly given that the improvement in the dominant term is significant. Homogenization complexity is $O(L \cdot m)$ ring-element multiplications, where $O(\cdot)$ hides a small constant (in our estimation settings, around 6). Multiplication (with modular reduction) is around $5\times$ more expensive than addition in our parameter regime. Also, unlike the projection

step, homogenization does not currently exploit short coefficients. Overall, homogenization may contribute roughly 30–40% of prover computation. These are preliminary estimates, and we leave a tighter analysis for future work.

3. We limited our discussion to the case where the cyclotomic ring is a power-of-two cyclotomic ring. This assumption simplifies the discussion of complex conjugation and automorphisms. However, the techniques can be extended to general cyclotomics by using a generalization of “constant term inspection” via algebraic trace and trace-dual bases, as shown in [KLNO25].
4. We limited our discussion to the ring variant of the R1CS relation. However, the techniques can be extended to the field variant of the R1CS relation with some additional technical overhead, as shown in [NS25]. Further, on an orthogonal note, the techniques can be extended to other relations, such as the CCS relation, as shown in [BC25a].

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